

Ghost Attack Modeling and Further Analysis

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I. GHOST TARGET SIMULATION

To evaluate robustness under adversarial conditions, synthetic ghost targets are injected into the measurement process. These ghost targets are designed to mimic realistic target dynamics by appearing and disappearing at different time steps and following the same motion and measurement models as the true targets.

Algorithm 1 summarizes the ghost-measurement generation procedure. At time step k , the ghost measurement set is initialized as

$$Z_k^{\text{ghost}} = \emptyset. \quad (1)$$

In a deception scenario, each ghost target $n = 1, \dots, N_{\text{ghost}}$ is defined by birth and death times $(t_{n,\text{birth}}, t_{n,\text{death}})$ and an initial state $x_{n,0}$. Its state evolves according to

$$x_{n,k} = \begin{cases} x_{n,0}, & k = t_{n,\text{birth}}, \\ \Phi(x_{n,k-1}), & k > t_{n,\text{birth}}, \end{cases} \quad (2)$$

where $\Phi(\cdot)$ denotes the same coordinated turn-rate motion model used for the true targets. The corresponding ghost measurement is generated as

$$z_k^{(n)} = h_k(x_{n,k}) + w_k, \quad w_k \sim \mathcal{N}(0, R_k), \quad (3)$$

where $h_k(\cdot)$ is the same measurement model used by the sensors. The generated measurement is then appended to Z_k^{ghost} .

In a replay scenario, previously recorded measurements are re-injected after a delay t_{rep} :

$$Z_k^{\text{ghost}} \leftarrow Z_k^{\text{ghost}} \cup Z_{\text{buf}}(k - t_{\text{rep}}), \quad k \geq t_{\text{rep}}, \quad (4)$$

where $Z_{\text{buf}}(\cdot)$ stores past measurements collected at the attacked node prior to replay.

In a flooding scenario, deception or replay is performed with a large number of ghost measurements in order to overwhelm the tracking process.

The complete ghost-target specification used in the simulations is reported in Table I. The nominal turn rate is defined as $\omega_{\text{turn}} = 2\pi/180$ rad/s, and individual ghost trajectories are obtained by scaling this nominal value. Some ghost targets share identical or closely spaced initial conditions and lifetimes; this is intentional and is used to emulate dense deception/flooding scenarios with multiple highly similar or duplicated tracks.

Algorithm 1 Ghost Attack Simulation at Time k

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1: Initialization:
2:    $Z_k^{\text{ghost}} \leftarrow \emptyset$ 
3:   choose scenario  $s \in \{\text{deception, replay, flooding}\}$ 
4:   if  $s = \text{deception}$  then
5:     for  $n = 1$  to  $N_{\text{ghost}}$  do
6:       initialize  $(t_{n,\text{birth}}, t_{n,\text{death}}, x_{n,0}, w_{n,\text{turn}})$ 
7:     end for
8:   else if  $s = \text{replay}$  then
9:     initialize replay buffer  $Z_{\text{buf}}$ 
10:    set replay delay  $t_{\text{rep}}$ 
11:   end if
12: Generate Ghost Measurements:
13: if  $s = \text{deception}$  then
14:   for  $n = 1$  to  $N_{\text{ghost}}$  do
15:     if  $t_{n,\text{birth}} \leq k \leq t_{n,\text{death}}$  then
16:       if  $k = t_{n,\text{birth}}$  then
17:          $x_{n,k} \leftarrow x_{n,0}$ 
18:       else
19:          $x_{n,k} \leftarrow \Phi(x_{n,k-1})$  (recursive propagation from previous
           time step)
20:       end if
21:       sample  $w_k \sim \mathcal{N}(0, R_k)$ 
22:        $z_k^{(n)} \leftarrow h_k(x_{n,k}) + w_k$ 
23:        $Z_k^{\text{ghost}} \leftarrow Z_k^{\text{ghost}} \cup \{z_k^{(n)}\}$ 
24:     end if
25:   end for
26: else if  $s = \text{replay}$  then
27:   if  $k \geq t_{\text{rep}}$  then
28:     for all  $\hat{z} \in Z_{\text{buf}}(k - t_{\text{rep}})$  do
29:        $Z_k^{\text{ghost}} \leftarrow Z_k^{\text{ghost}} \cup \{\hat{z}\}$ 
30:     end for
31:   end if
32: else if  $s = \text{flooding}$  then
33:   spawn ghosts as in deception or replay to simulate flooding
34: end if
35: Return  $Z_k^{\text{ghost}}$ 

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TABLE I

GHOST TARGET SPECIFICATIONS USED IN THE SIMULATION STUDY. FOR EACH GHOST TARGET, THE TABLE REPORTS BIRTH AND DEATH TIMES, INITIAL CARTESIAN POSITION, AND ASSIGNED TURN RATE.

ID	t_{birth}	t_{death}	x	y	Turn rate
1	10	100	1003	1488	$\omega_{\text{turn}}/8$
2	10	100	-255	1011	$-\omega_{\text{turn}}/3$
3	10	100	-1507	256	$-\omega_{\text{turn}}/2$
4	10	66	-1499	251	0
5	20	80	246	738	$\omega_{\text{turn}}/4$
6	20	80	246	738	$\omega_{\text{turn}}/4$
7	40	100	-242	993	$\omega_{\text{turn}}/2$
8	40	100	1000	1500	$\omega_{\text{turn}}/4$
9	40	80	250	750	$-\omega_{\text{turn}}/4$
10	40	100	-242	993	$\omega_{\text{turn}}/2$
11	40	100	1001	1501	$\omega_{\text{turn}}/4$
12	40	80	251	752	$-\omega_{\text{turn}}/4$
13	60	100	1000	1500	$-\omega_{\text{turn}}/4$
14	60	100	250	750	$\omega_{\text{turn}}/4$
15	60	100	999	1498	$-\omega_{\text{turn}}/4$
16	60	100	252	753	$\omega_{\text{turn}}/4$

II. ADAPTIVE ADVERSARIAL ATTACKS

To evaluate the robustness of KF-ALARM against informed adversaries, an adaptive attack scenario is considered. In this

setting, the adversary is assumed to have knowledge of the filtering mechanism and seeks to weaken the corroboration process by targeting nodes with limited connectivity and progressively compromising their neighbors. This reduces the availability of reliable cross-node support and degrades the effectiveness of the corroboration mechanism.

Node 6, which has a low degree (3), is selected as the primary target. The attack is then progressively intensified by expanding the set of compromised nodes as follows: (i) node 6 only; (ii) nodes 6 and 7; (iii) nodes 6, 7, and 2; and (iv) nodes 6, 7, 2, and 5. In each case, deception attacks with four ghost targets are applied by injecting measurements that mimic the motion of genuine targets.

TABLE II
KF-ALARM PERFORMANCE UNDER ADAPTIVE ATTACKS WITH PROGRESSIVELY COMPROMISED NODES (WINDOW=10, CUT-OFF=100 M, AND ORDER=1). LOWER VALUES INDICATE BETTER PERFORMANCE.

Metric	Compromised nodes			
	6	6,7	6,7,2	6,7,2,5
OSPA distance (m)	30.84	35.94	60.14	60.22
OSPA cardinality	7.83	14.28	49.19	49.28
OSPA ⁽²⁾ distance (m)	46.33	52.08	65.97	66.21
OSPA ⁽²⁾ cardinality	14.41	23.46	48.44	48.56

Table II summarizes the performance of KF-ALARM at the targeted node (6) under these adaptive attacks. When only node 6 is compromised, KF-ALARM maintains relatively stable tracking performance, indicating that corroboration from benign neighbors remains effective in suppressing adversarial measurements. As additional neighboring nodes are compromised, performance degrades progressively. In particular, the cardinality errors increase significantly when three or more nodes are attacked, reflecting the reduced availability of reliable corroboration and the increased influence of coordinated ghost measurements.

Notably, the degradation is gradual rather than abrupt, demonstrating that KF-ALARM retains partial robustness even under adversarial expansion. However, as the majority of neighboring nodes become compromised, the corroboration mechanism weakens, leading to increased false tracks and higher estimation error. These results are consistent with the expected behavior of consensus-based defenses under adaptive adversarial strategies.

III. BENIGN SCENARIO PERFORMANCE

This section reports performance under benign conditions, that is, in the absence of adversarial manipulation. The purpose is to verify that KF-ALARM preserves nominal tracking performance relative to the standard EKF baseline.

Performance is evaluated using the OSPA distance, the OSPA⁽²⁾ distance, and their associated cardinality components. These metrics quantify instantaneous multi-object error, trajectory consistency over a finite temporal window, and target-count accuracy, respectively.

Table III summarizes the average benign-case performance. KF-ALARM produces slightly larger distance errors. This behavior arises because the corroboration factor effectively inflates the measurement covariance when it is less than one,

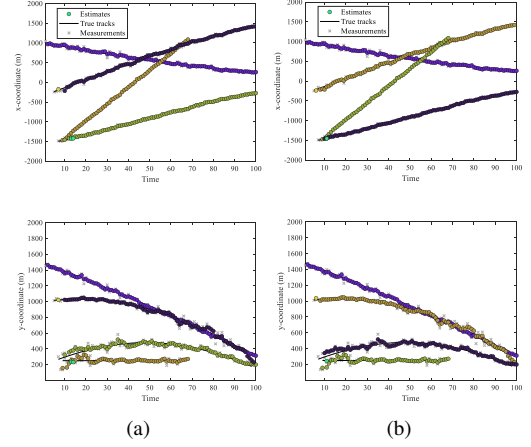


Fig. 1. Tracking performance under benign conditions: (a) KF-ALARM, and (b) the standard EKF.

reducing the Kalman gain and yielding more conservative updates. As a result, measurements are not fully trusted even in benign conditions, leading to a small loss in estimation accuracy. Fig. 1 illustrates the tracking performance of KF-ALARM and the standard EKF under benign conditions.

TABLE III
AVERAGE TRACKING PERFORMANCE IN THE BENIGN SCENARIO ($W = 10$, $c = 100$ M, $p = 1$). LOWER VALUES INDICATE BETTER PERFORMANCE. BEST RESULTS ARE SHOWN IN BOLD.

Metric	KF-ALARM	Standard
OSPA Distance Mean (m)	25.19	24.44
OSPA Cardinality Mean	6.98	5.78
OSPA ⁽²⁾ Distance Mean (m)	34.40	33.67
OSPA ⁽²⁾ Cardinality Mean	7	5.45

IV. DETAILED RESULTS UNDER ADVERSARIAL ATTACKS

This section provides additional results under deception, replay, and flooding attacks, summarized in Fig. 2. The figure reports temporal OSPA and OSPA⁽²⁾ distances together with the estimated target cardinality, enabling a detailed assessment of tracking accuracy and cardinality stability over time.

Across all attack scenarios, the standard EKF exhibits progressively larger tracking errors and noticeable cardinality inflation as the number or severity of adversarial measurements increases. In contrast, KF-ALARM maintains lower errors and more accurate cardinality estimates, demonstrating improved robustness to adversarial measurement corruption.

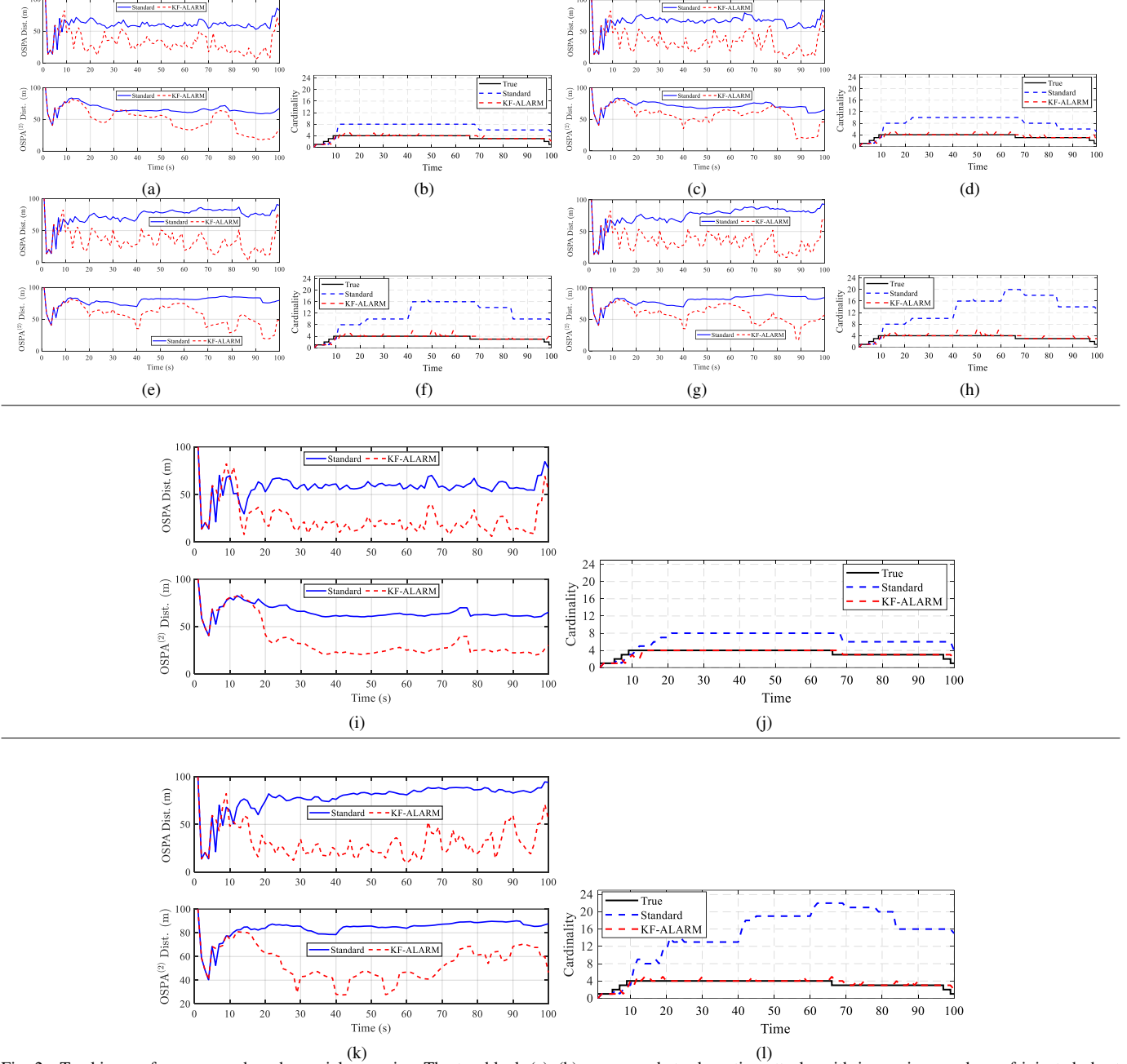


Fig. 2. Tracking performance under adversarial scenarios. The top block (a)–(h) corresponds to deception attacks with increasing numbers of injected ghost targets: (a)–(b) 4 ghosts, (c)–(d) 6 ghosts, (e)–(f) 12 ghosts, and (g)–(h) 16 ghosts. Within each pair, the left panel shows temporal OSPA and $OSPA^{(2)}$ distances, and the right panel shows the estimated and true target cardinalities. The middle block (i)–(j) shows the replay attack, and the bottom block (k)–(l) shows the flooding attack; in both cases, the left panel reports OSPA and $OSPA^{(2)}$, and the right panel reports cardinality.